solution! Figure 12.1 explains this concept: for some cluster shapes, our intuition tells us that the two rings should be in different clusters, but it's hard to come up with a metric function that objectively states so (minimum distance, for instance, wouldn't work). And this becomes even harder for high-dimensional datasets (where our intuition can't even help us validate these metrics, as it's hard to represent and interpret anything beyond 3-D spaces).

For these reasons, all clustering algorithms are heuristics that converge more or less quickly to a locally-optimal solution.

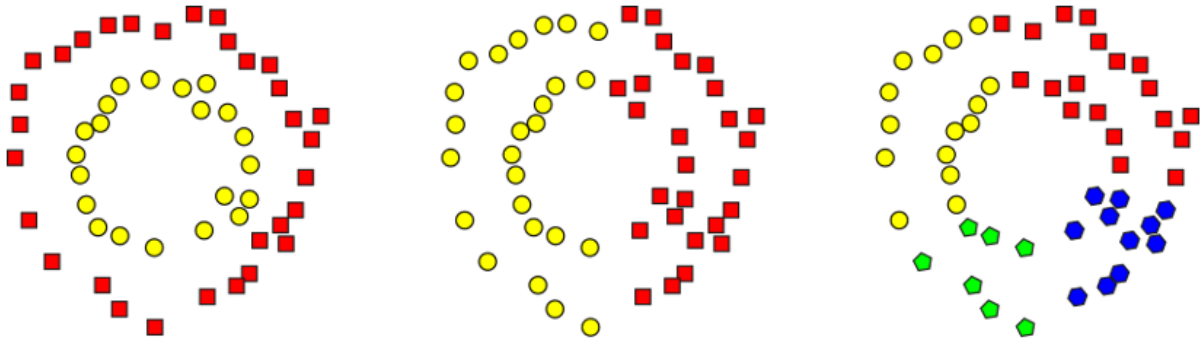


Figure 12.1 A dataset that's challenging to cluster. On the left, the ideal clustering, that matches our intuition. Next to it, non-optimal results produced by some metrics like proximity or affinity propagation.

Under the category labeled "clustering" we group several data-partitioning algorithms, using approaches completely different from each other; all these approaches can be applied to the problems described in the previous section almost transparently, although, obviously, each approach has strengths and flaws, and we should better select the best fitting algorithms based on our requirements.